

Math728J – Homework 1

1. Let $\mathbf{A} \in \mathbb{C}^{N \times N}$ and $\|\cdot\|$ be any matrix norm, prove that

$$\rho(\mathbf{A}) = \lim_{n \rightarrow \infty} \|\mathbf{A}^n\|^{1/n}.$$

2. Let $\mathbf{A} \in \mathbb{R}^{N \times N}$ and $\mathbf{A} = \mathbf{D} + \mathbf{L} + \mathbf{U}$ be the standard splitting. Show that if $\mathbf{A}$ is symmetric and positive definite, then $\mathbf{B}_{SGS} = (\mathbf{D} + \mathbf{L}^T)^{-1} \mathbf{D} (\mathbf{D} + \mathbf{L})^{-1}$ is also symmetric and positive definite.
3. Let $\mathbf{A} = (a_{ij}) \in \mathbb{R}^{N \times N}$ be an irreducibly diagonally dominant matrix with $a_{ii} > 0$ for $i = 1, \dots, N$. Show:
- For all eigenvalues $\lambda \in \mathbb{C}$ of $\mathbf{A}$ it holds that $\operatorname{Re} \lambda > 0$.
 - If $\mathbf{A}$ is also symmetric, then $\mathbf{A}$ is positive definite.
4. Consider an $N \times N$ tridiagonal matrix of the form $\mathbf{T}_\alpha = \operatorname{tridiag}\{-1, \alpha, -1\}$. where α is a real parameter.

- Verify that the eigenvalues of $\mathbf{T}_\alpha$ are given by

$$\lambda_j = \alpha - 2 \cos(j\theta), \quad j = 1, \dots, N,$$

where $\theta = \pi/(N+1)$ and an eigenvector associated with each λ_j is

$$v_j = [\sin(j\theta), \sin(2j\theta), \dots, \sin(nj\theta)]^T.$$

Under what condition on α does this matrix become positive definite?

- Now take $\alpha = 2$.
 - Will the Jacobi method converge for the matrix? If so, what will be its convergence factor?
 - Will the Gauss-Seidel method converge for the matrix? If so, what will be its convergence factor?
 - For which values of ω will the relaxation method converge?

5. For a given M-matrix $\mathbf{A} \in \mathbb{R}^{N \times N}$, prove the following estimate:

$$\rho(M_R(\omega_2)) \leq \rho(M_R(\omega_1)) < 1, \quad 0 < \omega_1 \leq \omega_2 \leq 1.$$

6. (**Programming Exercise**) For the numerical solution of the boundary value problem

$$\begin{cases} -u_{xx}(x, y) - u_{yy}(x, y) + b(x, y)u(x, y) = f(x, y), & (x, y) \in \Omega = (0, 1)^2 \\ u(x, y) = 0, & (x, y) \in \partial\Omega \end{cases} \quad (1)$$

where $b(x, y) = 1$ and $f(x, y) = (2\pi^2 + 1) \sin \pi x \sin \pi y$, we consider an uniform square grid with grid points (x_i, y_j) which are chosen to be equidistant:

$$x_i = ih, \quad y_j = jh, \quad i, j = 0, 1, \dots, N \quad (h = 1/N)$$

and the center difference scheme:

$$\frac{-U_{i-1,j} - U_{i,j-1} + 4U_{i,j} - U_{i+1,j} - U_{i,j+1}}{h^2} + b_{i,j}U_{i,j} = f_{i,j},$$

$$i, j = 1, 2, \dots, N-1,$$

where $b_{i,j} = b(x_i, y_j)$ and $f_{i,j} = f(x_i, y_j)$, and

$$U_{i,j} \approx u(x_i, y_j), \quad i, j = 1, 2, \dots, N-1$$

$$U_{i,0} = U_{i,N} = U_{0,i} = U_{N,i} = 0, \quad i, j = 0, 1, \dots, N.$$

For $N = 10, 20, 40$, determine an approximate solution of (1) using the Jaccobi method, Gauss-Seidel method and the Relaxation method with $\omega = 0.2, 0.4, 0.6, 0.8, 1.2, 1, 4, 1.6$, where the iteration is stopped when either more than 4000 iterations are reached or $\|U_k - U_{k-1}\|_\infty < 10^{-4}$. Choose $x_0 = 0$ as the initial value for the iterations. Hand in your codes with some explanations. For all methods, provide the number of required iteration steps k , $\|U_k - U_{k-1}\|_\infty$ and the errors $\max_{i,j=1,2,\dots,N-1} |U_{i,j} - u(x_i, y_j)|$ (you should be able to solve the true solution u yourself) in a table. You could use any Language such as C/C++, Fortran or Matlab.